# Gaussian Fourth Degree Function Report

## Polynomial

$$f(q) = aq^4 + bq^3 + cq^2 + dq + e$$

## Naive

$$g_{\text{naive}}(X) = X^4 a + X^3 b + X^2 c + Xd + e$$

$$\mathbb{E}[g_{\text{naive}}(X)] = aq^4 + 6aq^2\sigma^2 + 3a\sigma^4 + bq^3 + 3bq\sigma^2 + cq^2 + c\sigma^2 + dq + e$$

$$\begin{aligned}\text{Var}(g_{\text{naive}}(X)) = \sigma^2 \big(&16a^2q^6 + 168a^2q^4\sigma^2 + 384a^2q^2\sigma^4 + 96a^2\sigma^6 + 24abq^5 \\ &+ 168abq^3\sigma^2 + 192abq\sigma^4 + 16acq^4 + 72acq^2\sigma^2 + 24ac\sigma^4 \\ &+ 8adq^3 + 24adq\sigma^2 + 9b^2q^4 + 36b^2q^2\sigma^2 + 15b^2\sigma^4 + 12bcq^3 \\ &+ 24bcq\sigma^2 + 6bdq^2 + 6bd\sigma^2 + 4c^2q^2 + 2c^2\sigma^2 + 4cdq + d^2\big)\end{aligned}$$

$$\begin{aligned}\text{MSE}(g_{\text{naive}}) = \sigma^2 \big(&16a^2q^6 + 204a^2q^4\sigma^2 + 420a^2q^2\sigma^4 + 105a^2\sigma^6 + 24abq^5 \\ &+ 204abq^3\sigma^2 + 210abq\sigma^4 + 16acq^4 + 84acq^2\sigma^2 + 30ac\sigma^4 + 8adq^3 \\ &+ 24adq\sigma^2 + 9b^2q^4 + 45b^2q^2\sigma^2 + 15b^2\sigma^4 + 12bcq^3 + 30bcq\sigma^2 \\ &+ 6bdq^2 + 6bd\sigma^2 + 4c^2q^2 + 3c^2\sigma^2 + 4cdq + d^2\big)\end{aligned}$$

$$\text{Bias}(g_{\text{naive}}) = \sigma^2 \left(6aq^2 + 3a\sigma^2 + 3bq + c\right)$$

## Unbiased

$$g_{\text{unb}}(X) = X^4 a + X^3 b - 6X^2 a\sigma^2 + X^2 c - 3Xb\sigma^2 + Xd + 3a\sigma^4 - c\sigma^2 + e$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^4 + bq^3 + cq^2 + dq + e$$

$$\begin{aligned}\text{Var}(g_{\text{unb}}(X)) = \sigma^2 \big(&16a^2q^6 + 72a^2q^4\sigma^2 + 96a^2q^2\sigma^4 + 24a^2\sigma^6 + 24abq^5 + 72abq^3\sigma^2 \\ &+ 48abq\sigma^4 + 16acq^4 + 24acq^2\sigma^2 + 8adq^3 + 9b^2q^4 + 18b^2q^2\sigma^2 \\ &+ 6b^2\sigma^4 + 12bcq^3 + 12bcq\sigma^2 + 6bdq^2 + 4c^2q^2 + 2c^2\sigma^2 + 4cdq + d^2\big)\end{aligned}$$

$$\begin{aligned}\text{MSE}(g_{\text{unb}}) = \sigma^2 \big(&16a^2q^6 + 72a^2q^4\sigma^2 + 96a^2q^2\sigma^4 + 24a^2\sigma^6 + 24abq^5 + 72abq^3\sigma^2 \\ &+ 48abq\sigma^4 + 16acq^4 + 24acq^2\sigma^2 + 8adq^3 + 9b^2q^4 + 18b^2q^2\sigma^2 \\ &+ 6b^2\sigma^4 + 12bcq^3 + 12bcq\sigma^2 + 6bdq^2 + 4c^2q^2 + 2c^2\sigma^2 + 4cdq + d^2\big)\end{aligned}$$

## Comparison

$$\text{mean gap} = \sigma^2 \left(-6aq^2 - 3a\sigma^2 - 3bq - c\right)$$

$$\begin{aligned}\text{variance gap} = \sigma^4 \big(&-96a^2q^4 - 288a^2q^2\sigma^2 - 72a^2\sigma^4 - 96abq^3 - 144abq\sigma^2 - 48acq^2 \\ &- 24ac\sigma^2 - 24adq - 18b^2q^2 - 9b^2\sigma^2 - 12bcq - 6bd\big)\end{aligned}$$

$$\begin{aligned}\text{MSE gap} = \sigma^4 \big(&-132a^2q^4 - 324a^2q^2\sigma^2 - 81a^2\sigma^4 - 132abq^3 - 162abq\sigma^2 - 60acq^2 \\ &- 30ac\sigma^2 - 24adq - 27b^2q^2 - 9b^2\sigma^2 - 18bcq - 6bd - c^2\big)\end{aligned}$$

$$\operatorname{Bias}(g_{\text{naive}})^2 = \sigma^4 \left(6aq^2 + 3a\sigma^2 + 3bq + c\right)^2$$